PACK WITH LOVE CLUB

From Data to Impact

24-Lesson Project-Based Math Curriculum

Project-Based Math for Social Impact • 2026–2027 • Revised September 9, 2026 (v4.2)

Annual Challenge (Year 1): Choose, source, price, market, sell, and evaluate a ready-made product — a real resale business. The product (and with it its color, style, and packaging) is chosen by the class in Lesson 3; the order quantity and the price are set by the class's own survey data. Year 2 designs Pack with Love's own product.

Learning Path: Mission → Business → Product & Source → Research → Production → Sales → Analysis → Impact

Core Learning Cycle: Question → Prediction → Data → Math → Decision → Action → New Data → Reflection

Statistics Backbone: Pose a Statistical Question → Decide What to Measure → Collect Data → Summarize → Analyze → Interpret → *Decide & Act* (the course's added step). The class runs this cycle twice: once on the pricing survey (L4–L8, L12) and once on real sales (L18–L21).

Session structure: 24 sessions = 21 Sunday classes + 2 sales-event fieldwork days (Sales Event I ≈ Sat Feb 6, Sales Event II ≈ Sat Feb 20) + the Final Impact Showcase on March 14 (Session 24). Survey Week (≈ Oct 18–25) is homework fieldwork attached to Session 6. Order placed ≈ Nov 23 on the path the class chooses in L9 — a China supplier (sea or air) or a US wholesaler, weighed on equal footing; the shipment must arrive by Jan 10. Breaks: Nov 15, Nov 29, Dec 20, Dec 27; Feb 14 is a reserve Sunday.

AP Statistics column: each lesson lists the topics it touches from the **revised AP Statistics Course and Exam Description effective Fall 2026** (five units; first exam May 2027) — Unit 1 *Exploring One-Variable Data and Collecting Data*, Unit 2 *Probability, Random Variables, and Probability Distributions* — the CED skills it exercises (1.A determine an investigative question · 2.A identify information · 2.B justify an ethical collection method · 3.A construct representations · 3.B calculate summary statistics · 3.C estimate counts, percentages, probabilities · 4.A describe and compare · 4.B justify a claim · 4.C describe distributions · 4.D interpret results), and the Statistical Practices: P1 Formulate Questions · P2 Collect Data · P3 Analyze Data · P4 Interpret Results. Concept level only; inference skills (2.C–2.E, 3.D–3.E, 4.E–4.G) and Units 3–5 are out of scope.

Delivery: Sunday classes are online (Zoom); the two sales events — both at the same farmers market — and the Showcase are in person. Hands go to the homes: one sample is mailed to each family for L9; each family inspects and then sells its own 3–5 units between L14 and L17; inventory is distributed at Event I. Class time is used for what only a full class online can do — every student's simulation draws pooled into one distribution (L14, L17). Every lesson is planned to 50–58 minutes with a marked cut line for its last segment.

Class Workbook and rubric: all deliverables live in one shared workbook — one tab per session, one row or cell per student — from which each Student Portfolio is exported at season's end. Three criteria apply to every deliverable, self-checked ✓/■: *the conclusion is in context* (4.D) · *every claim has evidence* (4.B) · *every number has a source* (the L3 habit).

Decision Ledger: Fixed by the end of L3 — the product, its look and packaging, and an initial sourcing lean (class recommendation, confirmed by Adult Advisors). Decided by data — order quantity (L10), price (L12), demand forecast and venue (L15), the mid-campaign adjustment (L19).

Phase 1 · Understand the Business & Choose the Product (3 sessions)

No.	Topic	Driving Question	Math & Thinking	Class Task	Deliverable	AP Stats (2026 CED)
1	What Is a Business? — Fundamentals (Guest: Prof. Kevin Jialin Sun)	What has to be true for a business to exist — and to survive?	Customer · product · price · cost; Revenue – Cost = Profit; cost types (seed); risk	Prof. Sun walks a familiar local business through four boxes — customer, product, price, cost — then the one equation that governs everything after: Revenue (price × sold) – Cost = Profit. Risk defined as "what makes the equation come out wrong." Use of funds and who decides (Charter). Students sketch a business they have actually bought from, with best-guess numbers, and challenge a partner's numbers.	Business Sketch + 2 Business Questions per student	—

Phase 1 · Understand the Business & Choose the Product (3 sessions)

No.	Topic	Driving Question	Math & Thinking	Class Task	Deliverable	AP Stats (2026 CED)
2	How Products Move — Supply Chain, Framework and Reality (Guest: Tony Yao)	How does a product move from an idea to something a customer can actually receive — and what does the textbook not tell you?	Sequence, order of magnitude, MOQ, lead time, cost/time/risk trade-off	Label flip: a "US wholesale" item and a custom sample both read Made in China — domestic vs. import is about who does the importing. Tony walks the full chain on a real order he has run, then the three entry points: upstream (custom in China), downstream (US wholesaler), middle (stock + local value-add). Students map three lanes and mark cost, time, and risk points. Tony answers the Lesson 1 question cards.	Supply Chain Map + Business Risk List	—
3	Choose the Product, Find the Source — Feasibility Screen & Landed Cost (Guest: Joyce Xie)	Which of these products should we sell — and where in the world would we actually get it from?	Landed-cost estimation, percentages, rates, source reliability	Students shortlist three candidates from real retail field research. Joyce teaches sourcing on a live supplier-platform search (teacher's screen; no student accounts): search terms, reading a listing — price tiers, MOQ, sample cost, lead time, port, weight, verification — and why a listed price is not a quoted price. Teams fill a Sourcing Comparison Sheet, build a landed-cost estimate from their own numbers (product + air/sea freight + tariff + clearance), compare with a US wholesale price, and draft a supplier inquiry. The product's color, style, and packaging are fixed with it — sold as-is this year. Charter rule stated aloud: students search, compare, draft; adults send, quote, request samples, order.	Sourcing Comparison Sheet + Landed-Cost Estimate + Supplier Inquiry Draft + Product Decision Record	—

Phase 2 · Market Research & Statistics (5 sessions incl. survey-week fieldwork)

No.	Topic	Driving Question	Math & Thinking	Class Task	Deliverable	AP Stats (2026 CED)
4	Pose the Statistical Question — Who Are We Asking?	The product is decided — color, style, packaging and all. What is still open is the price. What exactly are we asking, about whom, and why can't we ask everyone?	Investigative question (three components); population, sample, census; observational units; variables (categorical vs. quantitative); parameter vs. statistic (named); prediction	Introduce the statistics backbone diagram — four practices, run twice this season. Contrast a factual question (our landed cost) with a statistical one (what would people pay? — answers vary). Write the investigative question with its three components: the variable(s) that guide collection, what we estimate, and the population the conclusion applies to; fix it — it does not change after the data come in. Components of a study: population (N), sample (n), census, and why a census is impossible; "in context" as the CED defines it. A survey is an observational study. Build the variable list — would you buy (categorical), most you'd pay (quantitative), buy at three price points, grade — and classify each. The principle: an error made while collecting data cannot be repaired by any later analysis. Each student writes a prediction in context. Wording and drafting move to L5.	Investigative Question Card + Variable List + Prediction Sheet	1.1 (1.1.A components; 1.1.B investigative question — taught here) · 1.2 Variables (units; categorical/quantitative; parameter vs. statistic named) · 1.10.A–B (three components; census) · 1.10.D.4 (survey) · skills 1.A, 2.A · P1, P2
5	Bias, Two Ways — Whom We Ask and How We Ask	If each of us asks 30–50 people at our own school, whom can the results represent — and how do we ask without pushing people toward our answer?	Simple random sample, stratified sample, convenience sample; bias as systematic error; voluntary response, undercoverage; question wording bias; generalization	What makes a sample represent the population: random vs. convenience (random-number demo); stratified sampling — each student's school is a stratum; bias defined as systematic error; voluntary response and undercoverage named. Honest generalization: our sample is not randomly selected, so results apply only to students similar to those surveyed. Then the second kind of bias: sort leading vs. neutral wording; rules — no adjectives, no suggested answer, one idea per question. Teams build Survey v1 from a skeleton by choosing between two wordings per item and justifying the choice. Permission round: each student reports from their Student Guide (teacher yes / club on track / studio yes / outside setting); an Adult Advisor sends a formal letter only where a school asks for one. Add the answered-by box (student / parent) so school and studio responses are analyzed as separate populations.	Sampling Plan + Survey v1 + Permission Status	1.11 Random Sampling (SRS, stratified) · 1.12 Sampling Problems (bias as systematic error; voluntary response, undercoverage, convenience; question wording bias) · 1.10.E (generalization) · skills 2.A, 2.B · P2

Phase 2 · Market Research & Statistics (5 sessions incl. survey-week fieldwork)

No.	Topic	Driving Question	Math & Thinking	Class Task	Deliverable	AP Stats (2026 CED)
6	Survey Launch Lab — fieldwork week	Will strangers answer the way we practiced — and what happens to the data if we only ask friends?	Simulation, quota, coding, interviewer effect, nonresponse, response bias	Simulate the friends-only bias on a spreadsheet school (friends say $8.10, the school says $6.40); finalize quota cards — mixed grades, genders, at most one-third friends. Pilot the survey in breakout pairs; the interviewer effect (two-tone recordings); asking rules — read exactly, don't nod, no mission talk; ethics — voluntary, anonymous; keep a refusal log (nonresponse is data). Code and tally pilot data; freeze Survey v2; fieldwork briefing. Homework (Survey Week, ≈ Oct 18–25): each student collects responses at their own school under the approved School Support Request, following school rules and the safety note.	Quota Cards + Pilot Data + Survey v2 + (due next week) Raw Data Set, 30–50 responses per student	2.3 Estimating Probabilities Using Simulation · 1.12 (nonresponse, response bias) · 1.10.E (ethical collection) · skills 2.A, 2.B, 3.C · P2
7	From Raw Data to Percentages — and to Averages	If 21 of 40 people say they'd pay $7, is that actually a lot?	Frequency, fraction, percentage; dotplot, mean, median, spread; five-number summary, boxplot (extension); sampling variability (intuition)	Class dataset merged by the teacher before class; students check it. Frequency table and percentages for categorical answers; a dotplot of "most you'd pay" with mean, median, and spread for the quantitative one; extension: five-number summary and boxplot. Parameter (the true population value, unknown) vs. statistic (our sample's value). Split the dataset by school and watch the proportions move — the first look at sampling variability. Check totals and errors. Predictions are opened next week.	Clean Data Table + Percentage & Distribution Summary	1.2.A (parameter vs. statistic) · 1.3 Categorical Tables · 1.5 Quantitative Graphs (dotplot) · 1.6 Describing Distributions · 1.7 Summary Statistics (mean, median, range) · 1.8 Five-Number Summary & Boxplot (extension) · 1.9 Comparing Distributions · 2.12 Sampling Distributions (intuition) · skills 3.A, 3.B, 4.A, 4.C · P3
8	Graphs, Evidence & the Price Signal	How can we make the market message understandable in 10 seconds?	Bar graph, two-way table (joint / marginal / conditional relative frequency), association, comparison	Open the L4 predictions against the data. Bar graph for categorical answers; two-way table (grade × buy at $7) with joint, marginal, and conditional relative frequencies — is grade associated with buying? Claim–Evidence–Reasoning states what the survey says about price and demand; the Price Signal is that CER's conclusion. AI checkpoint (8 min): let AI draft one chart, then find what it got wrong.	Market Research Brief + Preliminary Price Signal	1.4 Categorical Graphs · 1.9 Comparing Distributions · 2.1–2.2 Two Categorical Variables (joint, marginal, conditional relative frequency; association) · skills 3.A, 3.B, 4.A, 4.B, 4.D · P3, P4

Phase 3 · Sourcing, Order & Business Math (4 sessions; order placed after L10, ≈ Nov 23)

No.	Topic	Driving Question	Math & Thinking	Class Task	Deliverable	AP Stats (2026 CED)
9	Samples in Hand — Approve, Compare, and the Mathematics of a Weighted Decision	Is the cheapest supplier always the best supplier — and how much do the weights decide?	Weighted average as a mathematical object; percentage weights; inequality ("how much weight would flip the winner?"); evidence-based scoring	Business (20 min): one sample was mailed to each family — checklist against the listing and the L3 decision (size, color, packaging, safety label); quotes vs. L3 listing prices in one line (listed ≠ quoted, proven). Knock-out: can this supplier deliver by Jan 10 — sea with a committed ship date, air at a quoted price, or US stock? Criteria and weights were voted before class; debate only the contested ones; score China and US wholesale on equal footing, price as landed cost. Math (30 min): compute the weighted score by hand — Σ(weight × score) ÷ 100 — and why weights must total 100; then the flip question: how much weight must lead time carry before the US wholesaler wins? — an inequality in one unknown (7.EE.4). Recommend, with a robustness note. Jiangang Yao reviews.	Sample Inspection Record + Supplier Comparison Matrix + Flip-Point Note + Sourcing Recommendation	3.B (weighted calculation) · 7.EE.4 inequality (Common Core) — business decision mathematics
10	Cost per Unit, Quantity & the Order	What does one unit really cost, how many should we buy, and are we ready to spend real money?	Unit rate, fixed/variable cost, percent buffer, budget ceiling as a constraint	Name fixed vs. variable costs (seeded in L1) in a pre-built shared cost model; note what is absent this year — no tooling, no packaging design. Cost per unit at 200/300/500. Set order quantity from L8 demand data with an optimism haircut and a 10–15% buffer; check against the budget ceiling; go/no-go. Confirm the sourcing path the L9 matrix chose — China by sea or air, or US wholesale — against the Jan 10 arrival deadline; the quantity and budget arithmetic is the same on either path. Adult Advisors approve and place the order this week. (The MOQ trap and sensitivity analysis move to L11.)	Unit Cost Model + Production Approval Packet → order placed	—

Phase 3 · Sourcing, Order & Business Math (4 sessions; order placed after L10, ≈ Nov 23)

No.	Topic	Driving Question	Math & Thinking	Class Task	Deliverable	AP Stats (2026 CED)
11	Landed Cost & Import Math — the Real Numbers, and What We Didn't Buy	Why is factory price not the same as final cost — how wrong were we in September — and what if we had ordered 500?	Rates, percent, landed cost, percent error, sensitivity analysis	Rebuild the L3 chain with real numbers from the placed order — for a China order: quoted price, actual freight, current tariff rate, clearance; for a US wholesale order: wholesale price, domestic shipping, sales tax → landed cost per unit; compare with the path not taken. Percent error of the L3 estimate, line by line; reopen "the number I trusted least." Then the MOQ trap in hindsight: cost per unit at 500 was lower — what would the cash tied up in unsold units have been? Sensitivity: tariff ±10 points, freight doubled — one input at a time.	Domestic vs. Import Cost Scenario + Percent-Error Note	—
12	Pricing Scenarios — the Data Sets the Price	$5, $7, or $10 — which price is actually better?	Revenue, gross margin, scenario analysis, break-even, sell-through	Combine the L8 Price Signal with the real landed cost; model price/volume scenarios; discount stated willingness-to-pay for optimism; break-even and — since the order is placed — the sell-through share needed at each price. Recommend a price in CER form.	Pricing Recommendation	—

Phase 4 · Marketing, Arrival & Sales (8 sessions incl. home practice sale and 2 sales events, Feb 6 & Feb 20)

No.	Topic	Driving Question	Math & Thinking	Class Task	Deliverable	AP Stats (2026 CED)
13	Mission-Based Marketing	How can we sell the product without turning the mission into a slogan?	Message testing, experiment design	Write and test a value proposition that combines product value with the Pack with Love mission; honest product line: chosen and priced by our class using survey data. Design a two-message test with coin-flip assignment, run during the following week.	Core Message + Sales Pitch + Message Test Design	1.1.B applied (which message sells?) · 1.10.C (experiment: units, treatments, response) · 1.13.A, 1.13.D (random assignment → causal claim, for similar people) · skills 1.A, 2.A, 2.B · P1, P2

Phase 4 · Marketing, Arrival & Sales (8 sessions incl. home practice sale and 2 sales events, Feb 6 & Feb 20)

No.	Topic	Driving Question	Math & Thinking	Class Task	Deliverable	AP Stats (2026 CED)
14	The Boxes Are Open — Arrival Inspection, and What a Sample Can Tell You	Is what arrived what we approved — and if 2 of 40 fail, is the shipment bad?	Percent of a quantity; random sampling of units; simulation; sampling distribution of a sample proportion (intuition)	Business (15 min): the shipment is at the lead teacher's home. Count vs. order on camera; one-sentence defect definition against the approved sample; acceptable rate ≤ 3%; the teacher inspects 40 units chosen by student-called random numbers; each family inspects its own 3–5 units before L17 on the same log. Statistics (35 min): build a simulated shipment in the class workbook — 250 units, true defect rate 3% — and let each student draw 40; fifteen draws become a dotplot of "defects found": 0, 1, 1, 2, 0, 3… A 3% batch produces 2-of-40 routinely. Then the reverse: if the truth were 10%, how often would we see only 2? Decide the rule: a second sample if the first fails. Risk register to the sales dates.	Arrival Inspection Log + Defect Definition + Simulation Dotplot + Risk Register	1.2 (operational definition) · 1.11 (SRS of units) · 2.3 (simulation) · 2.12 (sampling distribution of a proportion, by simulation) · skills 2.A, 3.A, 3.C, 4.C · P2, P3
15	Forecast Demand & Choose the Market	How much should we bring — and which farmers market do we set up at, twice?	Forecast, random process, law of large numbers, expected value, weighted criteria (reused from L9)	Message-test results summarized by the teacher before class (one chart). Turn "how many will buy here?" into an investigable question: traffic × stop rate × buy rate; low/base/high by varying each rate; expected revenue = price × buyers, and why more traffic makes the forecast steadier (law of large numbers). Both sales events run at the same farmers market so that Event I → Event II compares like with like; the weighted matrix (L9's method) chooses which market — traffic, audience fit, permission, fees, date fit. Audience fit is the honest gap: the survey population is students, the market's buyers are adults — so design a 5-question adult mini-survey for the table's first 30 minutes. Outreach draft moves to L16; inventory split to L17.	Sales Forecast (low/base/high) + Market Ranking + Table Mini-Survey	1.1.B applied · 1.10.E (audience fit — to whom conclusions apply) · 2.3 (random process, law of large numbers) · 2.4 · 2.9 (expected value) · skills 1.A, 2.A, 3.C · P1, P3

Phase 4 · Marketing, Arrival & Sales (8 sessions incl. home practice sale and 2 sales events, Feb 6 & Feb 20)

No.	Topic	Driving Question	Math & Thinking	Class Task	Deliverable	AP Stats (2026 CED)
16	Sales System, Event Day Kit & Two Ways to Run an Experiment	What must we record on sales day so the event runs — and so that Event II can actually test something?	Data fields, conversion rate, average revenue; experiment design: whole-event comparison vs. randomized alternating slots	Design the tally sheet (30-minute slot · stopped · asked price · bought · units · price paid · age group · feedback code · A/B slot label) — "stopped" is the denominator for conversion; inventory log; cash rule: an adult holds all money; roles rotate every 30 minutes. Then the experiment, in CED terms: Design 1, whole-event comparison (Event I vs. Event II) — same market, but day, weather, and traffic differ; no random assignment, so no causal claim. Design 2, alternating slots within Event II — the adjustment switched on and off every 30 minutes in an order decided by coin flip; slots are the experimental units, assignment is random, so a causal claim is allowed for that market on that day. Students draft the market outreach ask; Adult Advisors book both dates.	Sales Tracker + Team Roles + Event Day Kit + Outreach Draft	1.10.C–D (experiment vs. observational; confounding) · 1.13.A (experimental units = slots; random assignment by coin flip; control; replication) · 1.13.D (random assignment → causal claim, scoped) · skills 2.A, 2.B · P2
17	Home Practice Sale — and Turning the Forecast into a Distribution	Does our system survive contact with real customers — and how much stock is "enough"?	Conversion, reconciliation; simulation; distribution of an outcome; expected value with variability; percentile-based decision	Business (20 min): each family sold 3–5 units at home during the week with the real tally sheet; pool the tallies, reconcile units × price against cash, compute conversion and revenue per buyer, find the two weakest points in the system and fix them → Kit v2. Statistics (30 min): each student simulates one Saturday in the class workbook — traffic, stop rate, and buy rate each drawn around the L15 estimates — and reports units sold; the class's simulated Saturdays become a dotplot. Low/base/high are no longer three guesses but the tails and middle of a distribution. Decide inventory for Event I from it: how many units give roughly an 80% chance of not selling out? Split the rest to Event II.	Rehearsal Tally + Reconciliation + Kit v2 + Forecast Distribution + Inventory Split	2.3 (simulation) · 2.9 (expected value with variability) · 2.12 (distribution of an outcome, by simulation) · skills 2.A, 3.A, 3.C, 4.C · P2, P3
18	Sales Event I — fieldwork (≈ Sat Feb 6)	Does real customer behavior match what people said in the survey?	Real-time tally, conversion, inventory	Run the first real sale using the Event Day Kit. Record inquiries, buyers, price, inventory change, and feedback.	Sales Data Set A	1.2 (observational units) · skill 2.A · P2

Phase 4 · Marketing, Arrival & Sales (8 sessions incl. home practice sale and 2 sales events, Feb 6 & Feb 20)

No.	Topic	Driving Question	Math & Thinking	Class Task	Deliverable	AP Stats (2026 CED)
19	Analyze & Adjust Mid-Campaign — Which Variables May Be Randomized?	Should we keep the plan, change the pitch, the sign, the bundle — or the price? And which of those can Event II test fairly?	Percentage change, prediction error by input, stated vs. revealed; ethics of random assignment	Compare forecast with actual: which input was off — traffic, stop rate, or buy rate? Survey said 52% would buy at $7; at the table 31% of those who stopped did — the stated-vs-revealed gap, and the adult mini-survey beside it. Tally feedback codes. Choose one adjustment, then classify it: pitch, sign, table layout, and bundles can be switched on and off by slot (Design 2); price cannot — the same customer must not meet two prices on the same day, so a price change can only be tested whole-event (Design 1). Some variables cannot ethically be randomized. Write the Event II question and a numeric prediction before the data — not to be changed after.	Mid-Campaign Decision Memo (question, design, prediction)	1.1.B student-generated (fixed before the data, 1.1.B.1) · 1.13.A (one treatment; which variables can be randomized — ethics) · 2.12 (intuition) · skills 1.A, 3.B, 4.B, 4.D · P1, P3, P4
20	Sales Event II / Campaign Close — fieldwork (≈ Sat Feb 20, same market)	What happened when we switched the change on and off?	Cumulative total, rate, comparison by slot	Same market, same recording system. If the adjustment is slot-switchable, run it by the coin-flip slot schedule (A/B labels on every tally sheet); if it is a price change, run the whole event on the new price. Full campaign reconciliation; unsold units counted.	Final Sales Data Set (slot-labeled)	1.13.A (randomized slot schedule executed) · skill 2.A · P2
21	Post-Sale Statistical Analysis	What did we predict correctly, where were we wrong — and did the change work?	Two-way table by slot (joint, marginal, conditional), percent change, segment analysis, stated vs. revealed, causal scope	Two-way table A-slots vs. B-slots: bought / didn't (of stopped) — the randomized comparison, and what it can claim (that market, that day). Event I vs. Event II whole-day comparison beside it, and what it cannot claim. Segments by age group and time slot; the campaign-wide stated-vs-revealed ratio for students and, from the mini-survey, for adults. AI checkpoint: AI may compute; every number is verified by hand before it enters the report. If the difference is small or inconclusive, say so — that is a finding.	Post-Sale Data Dashboard	2.1–2.2 (A/B two-way table) · 1.13.D (causal claim from the randomized slots; none from the whole-day comparison) · 1.10.D.5 · skills 3.A, 3.B, 4.A, 4.B, 4.D · P3, P4
22	Profit, Net Fundraising & Impact Math	Why is sales revenue not the same as the amount we truly raised?	Revenue, cost, net proceeds, margin	Calculate revenue, direct product costs, approved expenses, and net fundraising; convert net fundraising into the backpack-impact metric on the shared goal.	Financial & Impact Summary	—

Phase 4 · Marketing, Arrival & Sales (8 sessions incl. home practice sale and 2 sales events, Feb 6 & Feb 20)

No.	Topic	Driving Question	Math & Thinking	Class Task	Deliverable	AP Stats (2026 CED)
23	Business Review: What Would We Do Differently?	If we did this again, which decision would we change?	Evidence synthesis, trade-off	Review major decisions — product (L3), supplier (L9), quantity (L10), price (L12), venue (L15), mid-campaign change (L19); revisit each student's first-impression pick from L3; distinguish a good decision with a poor outcome from a poor decision with a lucky outcome. Assign Showcase storyboard beats.	Business Review Report + Showcase Storyboard	1.1.B reflected · skills 4.B, 4.D · P4
24	Final Impact Showcase — Sunday March 14, 2027 (rehearsal 4:00 PM, Showcase 5:00 PM; counts as Session 24)	Can we use evidence to explain the entire annual project clearly?	Data storytelling, visual communication	Present the mission, product choice, research, numbers, mistakes, improvements, and final impact to parents, community members, and supporters.	Final Presentation + Student Portfolio	skills 1.A–4.D as one narrative · P1–P4

Sourcing & the Student Charter

- Lesson 3 teaches students to find and read supplier listings on a global platform, using the teacher's screen and pre-verified listings; students hold no platform accounts.
- Students draft every supplier inquiry; the lead teacher sends them from the club's email (packwithloveclub@gmail.com), covering both international and Chinese-language platforms, requests samples, and records replies on the Sourcing Comparison Sheet for L9.
- Consistent with Charter §7 and §9: students search, read, compare, and draft; Adult Advisors contact suppliers, request quotes, approve the supplier (Jiangang Yao reviews at L9 and L10), and place the order after L10. No sample fees, deposits, or quantity commitments are made at the inquiry stage.

Fieldwork Design & School Partnership

Survey Week (≈ Oct 18–25) — distributed fieldwork

- Each student runs the class's pricing survey at their own school — or at a studio lobby, a church youth group, or another supervised setting — with permission they asked for themselves. Every student adds a stratum; school responses (students) and studio responses (parents) are marked and analyzed as separate populations. 30–50 responses each.
- Quota cards (from Lesson 5) require mixed grades, genders, and non-friends — the class's structural defense against friendship bias.
- Safety & ethics note for parents: surveys happen only at school among known peers or in supervised settings, following all school rules; no stranger outreach. Participation in the survey is voluntary and anonymous.

Asking for permission — the student guides, and the adult letter behind them

- Students ask for permission themselves, from the first week of the course, following the Bring It to Your School and Bring It to Your Studio student guides: one short email to a teacher (or one conversation with a studio owner), a club-chapter application where the school allows it, and a booth request in November. Hearing no is part of the skill. An Adult Advisor sends a formal School Support Request only where a school asks for a letter from an adult; the templates stand ready.
- An Adult Advisor reviews and submits every request to the teacher, counselor, or administration — consistent with the Student Charter: students draft and recommend; adults make official commitments.

- Each approval is recorded (school, contact, date, scope) by the Project Recorder; fundraising at any school follows that school's own fundraising rules.

Sales Events I & II (≈ Feb 6 and Feb 20, same farmers market) — designed, rehearsed, not improvised

- Market Plan (Lesson 15): candidate farmers markets scored on a weighted matrix (traffic, audience fit, permission, fees, date fit); both events run at the one chosen so that Event II can be compared with Event I. Students draft the outreach ask (Lesson 16); Adult Advisors make contact, handle fees or forms, and book both dates.
- Event Day Kit (Lesson 16): booth checklist · tally sheet · pitch card · price card · inventory log · customer feedback mini-cards — every student role has a sheet in hand; every transaction becomes a data point.
- Home Practice Sale (Lesson 17): each family sells 3–5 units at home with the real tally sheet — the data protocol is piloted, and the first real sales are made, before the events.
- The class's own experiment: Event I is the baseline; in Event II the chosen adjustment is switched on and off in 30-minute slots in a coin-flip order — a real randomized experiment on a real sales table. Students learn which adjustments can be randomized this way (pitch, sign, layout, bundle) and which cannot (price — the same customer must not meet two prices), and what each design can and cannot claim. An inconclusive result is reported as a finding, not hidden.
- Cash rule at every event: a supervising adult holds all money; students record, present, and sell. Net proceeds move through ACO's approved channel.
- Event dates are announced to families at least three weeks ahead; a student who cannot attend receives an alternative real role (pre-event prep or post-event analysis).

AP Statistics Connection (revised 2026 Course and Exam Description)

The College Board revised AP Statistics for the 2026–27 school year (first new exam May 2027): nine units became five, the four course skills became four **Statistical Practices** drawn from the ASA's GAISE framework — *Formulate Questions, Collect Data, Analyze Data, Interpret Results* — and the second-year-algebra prerequisite was removed. Those four practices are the four steps of this course's process diagram, run twice: on the pricing survey and on real sales. The course's statistics sit in the conceptual layer of the two units below. Terms are introduced after the experience they name, recorded on a running AP vocabulary card in each Student Portfolio, and never taught as computation: no inference formulas, no hypothesis tests, no regression (Units 3–5 are out of scope by design). Two AP habits of mind run throughout — every conclusion written *in context*, and every claim built as claim–evidence–reasoning. The new exam's first free-response question is devoted to Practices 1–2, which Sessions 4–6 train directly.

- **Unit 1 · Exploring One-Variable Data and Collecting Data** (20–30% of the exam) — 1.1–1.2 the investigative question with its three components (1.1.B taught in L4, applied in L13 and L15, student-generated in L19), observational units, variables, parameter vs. statistic (L4, L7); 1.10 population, sample, census, survey as observational study, experiment vs. observational study, confounding, and the rules for generalization (L4, L5, L16); 1.3–1.9 frequency tables, bar charts, dotplots, shape/center/spread, five-number summary and boxplot, comparing distributions (L7–L8) — all 13 Unit 1 topics touched at the concept level; 1.11 random and stratified sampling (L5, L14); 1.12 bias as systematic error — voluntary response, undercoverage, nonresponse, response and wording bias (L5–L6); 1.13 experimental design — random assignment and the causal claim it permits (L13), and why the sales experiment permits none (L16–L21).
- **Unit 2 · Probability, Random Variables, and Probability Distributions** — 2.1–2.2 two-way tables — joint, marginal, conditional relative frequency; association (L8, L21); 2.3 random process, simulation, law of large numbers (L6, L15); 2.4 probability as long-run relative frequency (L15); 2.9 expected value (L15); 2.12 sampling distributions — as intuition only, via proportions across schools, the arrival inspection, and prediction error (L7, L14, L21).

Why middle school students can do this: the same content is placed at the middle grades by the Common Core (6.SP.1–5, 7.SP.1–8, 8.SP.4) and by the American Statistical Association's GAISE framework (Level B), and the revised AP course now requires only first-year algebra. What this course adds is real data with real consequences, which makes the ideas more concrete, not harder.

The Long Game: Year 1 → Year 3 → Student-Led

One season is real. The track is longer. Each year runs the same four practices — Formulate Questions, Collect Data, Analyze Data, Interpret Results — on a project the students drive more, with statistics that deepen because the project demands it.

- **Year 1 · 2026–27 · A real resale business.** Choose a ready-made product, source it, price it by data, sell it as-is. Statistics: the concept layer of Unit 1 (all 13 topics) and the descriptive/simulation half of Unit 2 (2.1–2.4, 2.9, 2.12 as intuition). Skills 1.A, 2.A–2.B, 3.A–3.C, 4.A–4.D. No formulas, no inference.
- **Year 2 · 2027–28 · Design Pack with Love's own branded product.** The students' design, real tooling, packaging, minimum orders, a second selling season. Because they now control the product, the statistics grow where the project needs them: a real product-preference experiment with random assignment (1.13 in full — blocking, matched pairs), quality data on their own production run (1.7 standard deviation, IQR, outlier rules; 1.8–1.9), probability rules for defect and demand models (2.5–2.8), and expected value with variability (2.9). Two seasons of their own data become the material for the third year.
- **Year 3 + guided self-study · Toward the AP exam.** With two real seasons as the backbone, students complete the remaining two-thirds of the syllabus — the normal distribution and sampling distributions (2.10–2.12), inference for proportions and means (Units 3–4), regression (Unit 5) — through guided self-study or AP Statistics at school, and sit the exam. The revised course requires only first-year algebra.
- **Grade 10 and up · Student-led.** Under Student Charter §5.9, members grow into High School Student Mentors and run this course themselves — teaching Year 1 to the next cohort — with Adult Advisors holding the guardrails.

AP readiness depends on each student's pace. The course builds the thinking; the exam preparation on top is a manageable, guided step.

Program Parameters

- Recommended grades: Grades 6–8 (2026–27 cohort is primarily Grade 7); elementary students welcome with an accompanying parent (support, not substitution). Every lesson carries an upward extension for advanced learners.
- Core sessions: 24 sessions — 21 Sunday classes (≈60 minutes) + 2 sales-event fieldwork days + the Final Impact Showcase (Session 24). Survey Week is homework fieldwork attached to Session 6. Sundays 5:00 PM Pacific / 8:00 PM Eastern. No class Nov 15, Nov 29, Dec 20, Dec 27; Feb 14 held in reserve.
- Season: September 13, 2026 – March 7, 2027; Final Impact Showcase March 14, 2027. Milestones: product decision ≈ Sept 27 · Survey Week ≈ Oct 18–25 · pricing data ≈ Nov 1 · samples and quotes ≈ Nov 8 · order placed ≈ Nov 23 · price set ≈ Dec 13 · shipment arrives by Jan 10 · Sales Event I ≈ Feb 6 · Sales Event II ≈ Feb 20 · Business Review Mar 7. Dates are approximate; the real project comes first.
- Tuition: $30 per session; $720 for the 24-session season. Tuition pays for teaching and is not a donation.
- Class size: 6–15 students; the course runs with a minimum of 6. Roles rotate through Market, Product, Finance, and Sales; with a smaller class each student holds a role longer and the simulations (L14, L17) run more than one draw per student so the class dotplot has 12–20 points.
- Participation: Students do not need to travel to Africa. The full annual project is completed locally; the profit from class sales is the team's contribution to the Love & Hope Backpack Project's shared $5,300 goal (net of real product costs), which helps buy 1,000 color-it-yourself backpack sets for children across ten villages in Senegal.
- Final assessment: No traditional exam. Students complete a Student Portfolio, Business Review / Impact Report, and the Final Impact Showcase.
- Expectations, stated plainly: two sales events yield a few dozen transactions each — the class's experiment may well be inconclusive, and learning to say so is part of the course; the class's net proceeds are expected to be a modest share of the $5,300 goal and are always described as a team contribution.
- Continuity: each guest-led session has a 30-minute teacher-led backup plan so that no date moves if a mentor is unavailable; a parent logistics volunteer (no supplier contact) helps with unit distribution and event-day transport.
- AI use: AI may draft charts and compute — students must critique, verify, and take responsibility for every number. AI reduces unnecessary work, not necessary thinking.
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